

Preliminary,

The Zorn's Lemma gives several facts

- 1). Given a field F , there exists the Algebraic Closure $\overline{F}$.
- 2). Let F be a field and E/F is an Algebraic extension.
Given an Embedding $\varphi: F \rightarrow L$ where L is algebraically closed field,
there exists an Extension $\tilde{\varphi}: E \rightarrow L$ of φ .
Moreover, if E is algebraically closed, such $\tilde{\varphi}$ is Isomorphism.

In general, field Homomorphisms don't need to behave 'good' properties.

For instance, Let $K = \mathbb{Q}(\sqrt[3]{2})$. The minimal polynomial $x^3 - 2$ over $\mathbb{Q}$ of $\sqrt[3]{2}$ has three distinct roots, $\sqrt[3]{2}$, $\omega \cdot \sqrt[3]{2}$, $\omega^2 \cdot \sqrt[3]{2}$ where ω is primitive 3th root of unity. Therefore, $\# \text{Emb}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}), \mathbb{C}) = [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$.

But, since $\omega^i \sqrt[3]{2} \notin \mathbb{Q}(\sqrt[3]{2})$ $i=1, 2$, so

$$\text{Aut}(\mathbb{Q}(\sqrt[3]{2})) = \{\text{id}\},$$

